

PowerSys Calc

Comprehensive Validation Report

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Total calculators tested	29
Passed	28
Failed	1
Pass rate	96.6%

Scope: Every calculator in the project (29 across 9 categories) is exercised with hand-derived inputs. All outputs are validated — result rows (value, unit, status), raw values, recommended picks (cable size, OCPD, transformer kVA, etc.), overall status, summary, engineering notes / standards references, recommendations, warnings, formulas, and step-by-step derivations.

Test framework: tests/comprehensive-validation.mts — a standalone test runner that imports each calculator definition, invokes `compute()` with hand-derived inputs, and asserts on every output field.

Findings:

- `[transformer-efficiency]` status — expected "info", got "ok"

Summary — All Calculators

#	Category	Calculator	Test	Checks	Status
1	power	single-phase-power	120V · 15A · pf 0.85	36/36	PASS
2	three-phase	three-phase-power	480V · 100A · pf 0.8 (from V&I;)	35/35	PASS
3	power	power-factor	P=50kW · S=62.5kVA	21/21	PASS
4	power	pf-correction	100kW · 0.8→0.95 · 480V 3φ	30/30	PASS
5	conversion	kw-kva-hp	75 kW · pf 0.9	30/30	PASS
6	power	current-calculator	3φ 690V · 75kW · pf 0.85 · eff 0.95	21/21	PASS
7	power	demand-load	3 load groups · 5000h/yr	32/32	PASS
8	three-phase	delta-wye	Y · V · L→P · 690V	23/23	PASS
9	three-phase	balanced-load	Y · 690V · Z=20Ω · φ=45°	39/39	PASS
10	transformer	transformer-sizing	250kW · 480V · 3φ · df 0.9 · grow 15%	38/38	PASS
11	transformer	transformer-current	1000kVA · 22kV/400V · Z=5.75% · 3φ	38/38	PASS
12	transformer	transformer-efficiency	500kVA · P0=0.6 · Pk=5.5 · 80% load	32/33	FAIL
13	motor	motor-fl-current	22kW · 400V · 3φ · pf 0.85 · eff 0.9 (calc)	33/33	PASS
14	motor	motor-starting	45A · LR×7 · DOL · 400V · 30MVA	37/37	PASS
15	motor	vfd-sizing	45A · 400V · CT · 45°C · 1500m · 6kHz	36/36	PASS
16	motor	motor-breaker	45A · inverse-time · 3P	42/42	PASS
17	cable	voltage-drop	400V · 3φ · 80A · 60m · 16mm ² Cu · 60°C	39/39	PASS
18	cable	cable-sizing	100A · XLPE · C · Cu · 35°C · 2 circuits	47/47	PASS
19	cable	cable-derating	35mm ² PVC · C · 45°C · 3 circuits · soil 1.5	24/24	PASS
20	cable	short-circuit	25kA · 35mm ² Cu/XLPE · 0.3s · device I ² t=200M (FAIL)	32/32	PASS
21	protection	breaker-sizing	120A cont + 30A non · LV	33/33	PASS
22	protection	fuse-sizing	40A cont + 15A non · fast-acting	34/34	PASS
23	grounding	earthing-conductor	6000A · 0.3s · Cu/XLPE	28/28	PASS
24	grounding	ground-resistance	Single rod · ρ=50 · L=3m	32/32	PASS
25	renewable	solar-inverter	12kWp · DC/AC 1.15	28/28	PASS
26	renewable	battery-runtime	10kWh · 1.5kW · 80% DoD · 90% eff	28/28	PASS
27	renewable	solar-string	Vmp=41 · Voc=49 · -10°C to 70°C	36/36	PASS
28	conversion	hp-kw-converter	10 HP → kW	17/17	PASS
29	conversion	awg-mm2-converter	AWG 4 → mm ²	21/21	PASS

Single-Phase Power Calculator — single-phase-power

Test: 120V · 15A · pf 0.85 · Category: power · Result: **PASS** (36/36 checks)

Input

```
{
  "V": 120,
  "I": 15,
  "pf": 0.85,
  "phiUnit": "lagging"
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Apparent power S".value	1800	1800	✓
row:"Apparent power S".unit	"VA"	"VA"	✓
row:"Real power P".value	1530	1530	✓
row:"Real power P".unit	"W"	"W"	✓
row:"Reactive power Q".value	948.2	948.21	✓
row:"Reactive power Q".unit	"var"	"var"	✓
row:"Phase angle".value	31.79	31.79	✓
row:"Phase angle".unit	"\u00b0"	"\u00b0"	✓
raw.S	1800	1800	✓
raw.P	1530	1530	✓
raw.Q	948.2	948.2088377567467	✓
raw.phi	31.79	31.78833061705162	✓
status	"ok"	"ok"	✓
summary	"S = 1.8 kVA \u00b7 P = 1.53 kW \u00b7 Q = 0.948 kVAr"	"S = 1.8 kVA \u00b7 P = 1.53 kW \u00b7 Q = 0.948 kVAr"	✓
notes.count	2	2	✓
notes[0].standard	"IEEE"	"IEEE"	✓
notes[0].reference	"IEEE 1459-2010"	"IEEE 1459-2010"	✓
notes[0].text	"Defines measurement of AC electric power under sinusoida..."	"Defines measurement of AC electric power under sinusoida..."	✓
notes[1].standard	"IEC"	"IEC"	✓
notes[1].reference	"IEC 60038"	"IEC 60038"	✓
notes[1].text	"Standard nominal voltages: 230 V / 400 V (50 Hz) for IEC..."	"Standard nominal voltages: 230 V / 400 V (50 Hz) for IEC..."	✓
recommendations.count	0	0	✓
formulas.count	4	4	✓
formulas[0].name	"Apparent power"	"Apparent power"	✓
formulas[0].expression	"S = V \u00b7 I"	"S = V \u00b7 I"	✓

Field	Expected	Actual	Pass
formulas[0].variables	"V (V), I (A) \u2192 VA"	"V (V), I (A) \u2192 VA"	✓
formulas[1].name	"Real power"	"Real power"	✓
formulas[1].expression	"P = V \u00b7 I \u00b7 cos \u03c6"	"P = V \u00b7 I \u00b7 cos \u03c6"	✓
formulas[1].variables	"cos \u03c6 = power factor"	"cos \u03c6 = power factor"	✓
formulas[2].name	"Reactive power"	"Reactive power"	✓
formulas[2].expression	"Q = V \u00b7 I \u00b7 sin \u03c6"	"Q = V \u00b7 I \u00b7 sin \u03c6"	✓
formulas[2].variables	"sin \u03c6 = \u221a(1 \u2212 cos\u00b2\u03c6)"	"sin \u03c6 = \u221a(1 \u2212 cos\u00b2\u03c6)"	✓
formulas[3].name	"Power triangle"	"Power triangle"	✓
formulas[3].expression	"S\u00b2 = P\u00b2 + Q\u00b2"	"S\u00b2 = P\u00b2 + Q\u00b2"	✓
formulas[3].variables	"P, Q, S form a right triangle"	"P, Q, S form a right triangle"	✓
steps.count	">=4"	4	✓

Three-Phase Power Calculator — three-phase-power

Test: 480V · 100A · pf 0.8 (from V&I;) · Category: three-phase · Result: **PASS** (35/35 checks)

Input

```
{
  "Vll": 480,
  "I": 100,
  "pf": 0.8,
  "mode": "from_IV",
  "P_or_S": 30
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Apparent power S".value	83.138	83.138	✓
row:"Apparent power S".unit	"kVA"	"kVA"	✓
row:"Real power P".value	66.511	66.511	✓
row:"Real power P".unit	"kW"	"kW"	✓
row:"Reactive power Q".value	49.883	49.883	✓
row:"Reactive power Q".unit	"kVAr"	"kVAr"	✓
row:"Line current I".value	100	100	✓
row:"Line current I".unit	"A"	"A"	✓
row:"Phase voltage V_LN".value	277.13	277.1	✓
row:"Phase voltage V_LN".unit	"V"	"V"	✓
status	"ok"	"ok"	✓
summary	"P = 66.51 kW \u00b7 S = 83.14 kVA \u00b7 I = 100 A"	"P = 66.51 kW \u00b7 S = 83.14 kVA \u00b7 I = 100 A"	✓
notes.count	2	2	✓
notes[0].standard	"IEC"	"IEC"	✓
notes[0].reference	"IEC 60038"	"IEC 60038"	✓
notes[0].text	"Standard 3\u03c6 voltages: 400 V, 690 V (LV); 3.3, 6.6, ..."	"Standard 3\u03c6 voltages: 400 V, 690 V (LV); 3.3, 6.6, ..."	✓
notes[1].standard	"ANSI"	"ANSI"	✓
notes[1].reference	"ANSI C84.1"	"ANSI C84.1"	✓
notes[1].text	"Standard 3\u03c6 voltages: 208, 240, 480, 600 V (LV); 4...."	"Standard 3\u03c6 voltages: 208, 240, 480, 600 V (LV); 4...."	✓
recommendations.count	1	1	✓
recs[0]	"PF below 0.85 \u2014 consider capacitor bank for PF corr..."	"PF below 0.85 \u2014 consider capacitor bank for PF corr..."	✓
formulas.count	4	4	✓
formulas[0].name	"Apparent power"	"Apparent power"	✓

Field	Expected	Actual	Pass
formulas[0].expression	"S = \u221a3 \u00b7 V_LL \u00b7 I"	"S = \u221a3 \u00b7 V_LL \u00b7 I"	✓
formulas[0].variables	"V_LL (V), I (A) \u2192 VA"	"V_LL (V), I (A) \u2192 VA"	✓
formulas[1].name	"Real power"	"Real power"	✓
formulas[1].expression	"P = \u221a3 \u00b7 V_LL \u00b7 I \u00b7 cos \u03c6"	"P = \u221a3 \u00b7 V_LL \u00b7 I \u00b7 cos \u03c6"	✓
formulas[1].variables	"cos \u03c6 = power factor"	"cos \u03c6 = power factor"	✓
formulas[2].name	"Reactive power"	"Reactive power"	✓
formulas[2].expression	"Q = \u221a3 \u00b7 V_LL \u00b7 I \u00b7 sin \u03c6"	"Q = \u221a3 \u00b7 V_LL \u00b7 I \u00b7 sin \u03c6"	✓
formulas[2].variables	"sin \u03c6 = \u221a(1 - \u2212 cos\u00b2\u03c6)"	"sin \u03c6 = \u221a(1 - \u2212 cos\u00b2\u03c6)"	✓
formulas[3].name	"Line current"	"Line current"	✓
formulas[3].expression	"I = S / (\u221a3 \u00b7 V_LL)"	"I = S / (\u221a3 \u00b7 V_LL)"	✓
formulas[3].variables	"From S and V_LL"	"From S and V_LL"	✓
steps.count	">=4"	4	✓

Power Factor Calculator — power-factor

Test: P=50kW · S=62.5kVA · Category: power · Result: **PASS** (21/21 checks)

Input

```
{
  "mode": "from_PS",
  "P": 50,
  "S": 62.5,
  "Q": 30,
  "phi": 30
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Power factor (cos ϕ)".value	0.8	0.8	✓
row:"Phase angle ϕ ".value	36.87	36.87	✓
status	"warn"	"warn"	✓
summary	"PF = 0.8 (lagging) \u00b7 \u03c6 = 36.87\u00b0"	"PF = 0.8 (lagging) \u00b7 \u03c6 = 36.87\u00b0"	✓
notes.count	1	1	✓
notes[0].standard	"IEEE"	"IEEE"	✓
notes[0].reference	"IEEE 1459-2010"	"IEEE 1459-2010"	✓
notes[0].text	"For distorted waveforms, true PF = (P\u2081 / S) where P...	"For distorted waveforms, true PF = (P\u2081 / S) where P...	✓
recommendations.count	1	1	✓
recs[0]	"PF below 0.85 \u2014 consider installing capacitor bank ...	"PF below 0.85 \u2014 consider installing capacitor bank ...	✓
formulas.count	3	3	✓
formulas[0].name	"True PF"	"True PF"	✓
formulas[0].expression	"cos \u03c6 = P / S"	"cos \u03c6 = P / S"	✓
formulas[0].variables	"P (W), S (VA)"	"P (W), S (VA)"	✓
formulas[1].name	"From P & Q"	"From P & Q"	✓
formulas[1].expression	"cos \u03c6 = P / \u221a(P\u00b2 + Q\u00b2)"	"cos \u03c6 = P / \u221a(P\u00b2 + Q\u00b2)"	✓
formulas[1].variables	"P, Q"	"P, Q"	✓
formulas[2].name	"Phase angle"	"Phase angle"	✓
formulas[2].expression	"\u03c6 = cos\u207b\u00b9(PF)"	"\u03c6 = cos\u207b\u00b9(PF)"	✓
formulas[2].variables	"PF"	"PF"	✓
steps.count	">=0"	0	✓

Power Factor Correction Capacitor Sizing — pf-correction

Test: 100kW · 0.8→0.95 · 480V 3φ · Category: power · Result: **PASS** (30/30 checks)

Input

```
{
  "P": 100,
  "pf1": 0.8,
  "pf2": 0.95,
  "V": 480,
  "ph": "3"
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Capacitor bank Qc".value	42.13	42.13	✓
row:"Capacitor bank Qc".unit	"kVAr"	"kVAr"	✓
row:"Current before I■".value	150.33	150.4	✓
row:"Current before I■".unit	"A"	"A"	✓
row:"Current after I■".value	126.58	126.6	✓
row:"Current after I■".unit	"A"	"A"	✓
row:"Released capacity".value	15.8	15.8	✓
row:"Released capacity".unit	"%"	"%"	✓
picks.capacitorKVAR	42.13	42.13	✓
status	"err"	"err"	✓
summary	"Install \u2248 42.13 kVAr of capacitors to bring PF from..."	"Install \u2248 42.13 kVAr of capacitors to bring PF from..."	✓
notes.count	2	2	✓
notes[0].standard	"IEEE"	"IEEE"	✓
notes[0].reference	"IEEE 18-2002"	"IEEE 18-2002"	✓
notes[0].text	"Standard for shunt power capacitors. Capacitors are typi..."	"Standard for shunt power capacitors. Capacitors are typi..."	✓
notes[1].standard	"IEC"	"IEC"	✓
notes[1].reference	"IEC 60831-1/2"	"IEC 60831-1/2"	✓
notes[1].text	"Shunt power capacitors for AC systems having a rated vol..."	"Shunt power capacitors for AC systems having a rated vol..."	✓
recommendations.count	3	3	✓
recs[0]	"Round capacitor bank to nearest standard kVAr (typical 5..."	"Round capacitor bank to nearest standard kVAr (typical 5..."	✓
recs[1]	"If harmonic distortion is present (THD-V > 5%, THD-I > 8..."	"If harmonic distortion is present (THD-V > 5%, THD-I > 8..."	✓

Field	Expected	Actual	Pass
recs[2]	"Verify capacitor inrush current rating with switching co..."	"Verify capacitor inrush current rating with switching co..."	✓
formulas.count	2	2	✓
formulas[0].name	"Capacitor kVAr"	"Capacitor kVAr"	✓
formulas[0].expression	" $Q_c = P \tan^{-1} \left(\frac{\tan \phi_{load}}{\tan \phi_{capacitor}} \right)$ "	" $Q_c = P \tan^{-1} \left(\frac{\tan \phi_{load}}{\tan \phi_{capacitor}} \right)$ "	✓
formulas[0].variables	"P (W), ϕ_{load} = acos(pf _{load}), $\phi_{capacitor}$ = aco..."	"P (W), ϕ_{load} = acos(pf _{load}), $\phi_{capacitor}$ = aco..."	✓
formulas[1].name	"Released current"	"Released current"	✓
formulas[1].expression	" $I_{release} = I_{load} \sqrt{1 - \cos^2 \phi_{load}}$ "	" $I_{release} = I_{load} \sqrt{1 - \cos^2 \phi_{load}}$ "	✓
formulas[1].variables	"Reduction in line current"	"Reduction in line current"	✓
steps.count	">=3"	3	✓

kW / kVA / HP Converter — kw-kva-hp

Test: 75 kW · pf 0.9 · Category: conversion · Result: **PASS** (30/30 checks)

Input

```
{
  "value": 75,
  "pf": 0.9
}
```

Validation checks

Field	Expected	Actual	Pass
row:"W".value	75000	75000	✓
row:"W".unit	"W"	"W"	✓
row:"kW".value	75	75	✓
row:"kW".unit	"kW"	"kW"	✓
row:"kVA".value	83.333	83.333	✓
row:"kVA".unit	"kVA"	"kVA"	✓
row:"HP (mechanical)".value	100.57	100.577	✓
row:"HP (mechanical)".unit	"hp"	"hp"	✓
row:"PS (metric)".value	101.97	101.972	✓
row:"PS (metric)".unit	"PS"	"PS"	✓
row:"VAr".value	36323	36324	✓
row:"VAr".unit	"var"	"var"	✓
status	"ok"	"ok"	✓
summary	"75 kW = 100.58 HP = 83.33 kVA @ PF 0.9"	"75 kW = 100.58 HP = 83.33 kVA @ PF 0.9"	✓
notes.count	1	1	✓
notes[0].standard	"IEC"	"IEC"	✓
notes[0].reference	"ISO 80000-4"	"ISO 80000-4"	✓
notes[0].text	"1 mechanical HP = 745.6999 W. 1 metric HP (PS) = 735.498..."	"1 mechanical HP = 745.6999 W. 1 metric HP (PS) = 735.498..."	✓
recommendations.count	0	0	✓
formulas.count	3	3	✓
formulas[0].name	"kW \u2192 kVA"	"kW \u2192 kVA"	✓
formulas[0].expression	"kVA = kW / cos \u03c6"	"kVA = kW / cos \u03c6"	✓
formulas[0].variables	"cos \u03c6 = power factor"	"cos \u03c6 = power factor"	✓
formulas[1].name	"kW \u2192 HP"	"kW \u2192 HP"	✓
formulas[1].expression	"HP = kW \u00b7 1000 / 745.7"	"HP = kW \u00b7 1000 / 745.7"	✓
formulas[1].variables	"Mechanical horsepower"	"Mechanical horsepower"	✓
formulas[2].name	"kW \u2192 PS"	"kW \u2192 PS"	✓

Field	Expected	Actual	Pass
formulas[2].expression	"PS = kW \u00b7 1000 / 735.5"	"PS = kW \u00b7 1000 / 735.5"	✓
formulas[2].variables	"Metric HP (DIN 66036)"	"Metric HP (DIN 66036)"	✓
steps.count	">=0"	0	✓

Current Calculator (1ϕ / 3ϕ) — current-calculator

Test: 3ϕ 690V · 75kW · pf 0.85 · eff 0.95 · Category: power · Result: **PASS** (21/21 checks)

Input

```
{
  "phase": "3",
  "v": 690,
  "p": 75,
  "pf": 0.85,
  "eff": 0.95
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Apparent power S".value	92.88	92.879	✓
row:"Apparent power S".unit	"kVA"	"kVA"	✓
row:"Line current I".value	77.72	77.72	✓
row:"Line current I".unit	"A"	"A"	✓
status	"ok"	"ok"	✓
summary	"I = 77.72 A (3ϕ 690 V)"	"I = 77.72 A (3ϕ 690 V)"	✓
notes.count	1	1	✓
notes[0].standard	"IEC"	"IEC"	✓
notes[0].reference	"IEC 60364"	"IEC 60364"	✓
notes[0].text	"Design current is the current the circuit is intended to..."	"Design current is the current the circuit is intended to..."	✓
recommendations.count	2	2	✓
recs[0]	"Apply 125% factor for continuous loads (>3h): I_cont = 9..."	"Apply 125% factor for continuous loads (>3h): I_cont = 9..."	✓
recs[1]	"Select OCPD next standard size up."	"Select OCPD next standard size up."	✓
formulas.count	2	2	✓
formulas[0].name	"3ϕ current"	"3ϕ current"	✓
formulas[0].expression	"I = P / (√3 V_LL PF)"	"I = P / (√3 V_LL PF)"	✓
formulas[0].variables	"P (W), V_LL, PF, "	"P (W), V_LL, PF, "	✓
formulas[1].name	"1ϕ current"	"1ϕ current"	✓
formulas[1].expression	"I = P / (V PF)"	"I = P / (V PF)"	✓
formulas[1].variables	"P (W), V, PF, "	"P (W), V, PF, "	✓
steps.count	">=0"	0	✓

Demand Load & Load Factor Calculator — demand-load

Test: 3 load groups · 5000h/yr · Category: power · Result: **PASS** (32/32 checks)

Input

```
{
  "loads": "[{\\"name\\":\\"Lighting\\",\\"qty\\":80,\\"unit\\":0.05,\\"df\\":0.9},{\\"name\\":\\"HVAC\\",\\"qty\\":3,\\"unit\\":10,\\"df\\":0.7},{\\"name\\":\\"Receptacles\\",\\"qty\\":50,\\"unit\\":0.18,\\"df\\":0.4}]",
  "hoursPerYear": 5000
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Connected load".value	43	43	✓
row:"Connected load".unit	"kW"	"kW"	✓
row:"Maximum demand".value	28.2	28.2	✓
row:"Maximum demand".unit	"kW"	"kW"	✓
row:"Annual energy".value	141	141	✓
row:"Annual energy".unit	"MWh"	"MWh"	✓
row:"Average demand".value	16.1	16.1	✓
row:"Average demand".unit	"kW"	"kW"	✓
row:"Load factor".value	57.1	57.1	✓
row:"Load factor".unit	"%"	"%"	✓
status	"ok"	"ok"	✓
summary	"Peak demand 28.2 kW from 43 kW connected (3 groups)"	"Peak demand 28.2 kW from 43 kW connected (3 groups)"	✓
notes.count	2	2	✓
notes[0].standard	"NEC"	"NEC"	✓
notes[0].reference	"NEC 220"	"NEC 220"	✓
notes[0].text	"Demand factors per NEC Table 220.42 (lighting), 220.44 (..."	"Demand factors per NEC Table 220.42 (lighting), 220.44 (..."	✓
notes[1].standard	"IEC"	"IEC"	✓
notes[1].reference	"IEC 60364"	"IEC 60364"	✓
notes[1].text	"Demand factors per IEC 60364-8-1 (low-voltage installati..."	"Demand factors per IEC 60364-8-1 (low-voltage installati..."	✓
recommendations.count	1	1	✓
recs[0]	"Validate demand factors against local code \u2014 NEC an..."	"Validate demand factors against local code \u2014 NEC an..."	✓
formulas.count	3	3	✓
formulas[0].name	"Connected load"	"Connected load"	✓
formulas[0].expression	"\u03a3(qty \u00b7 unit)"	"\u03a3(qty \u00b7 unit)"	✓

Field	Expected	Actual	Pass
formulas[0].variables	"Per load group"	"Per load group"	✓
formulas[1].name	"Maximum demand"	"Maximum demand"	✓
formulas[1].expression	"\u03a3(qty \u00b7 unit \u00b7 DF)"	"\u03a3(qty \u00b7 unit \u00b7 DF)"	✓
formulas[1].variables	"DF = demand factor (NEC 220)"	"DF = demand factor (NEC 220)"	✓
formulas[2].name	"Load factor"	"Load factor"	✓
formulas[2].expression	"LF = avg / peak"	"LF = avg / peak"	✓
formulas[2].variables	"avg = annual energy / 8760 h"	"avg = annual energy / 8760 h"	✓
steps.count	">=0"	0	✓

Delta ↔ Wye Voltage & Current Converter — delta-wye

Test: Y · V · L→P · 690V · Category: three-phase · Result: PASS (23/23 checks)

Input

```
{
  "system": "wye",
  "quantity": "V",
  "direction": "L2P",
  "value": 690
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Input".value	690	690	✓
row:"Input".unit	"V"	"V"	✓
row:"phase".value	398.37	398.37	✓
row:"phase".unit	"V"	"V"	✓
status	"ok"	"ok"	✓
summary	"690 V \u2192 phase = 398.37 V (Y (wye))"	"690 V \u2192 phase = 398.37 V (Y (wye))"	✓
notes.count	1	1	✓
notes[0].standard	"IEEE"	"IEEE"	✓
notes[0].reference	"IEEE Std 141 (Red Book)"	"IEEE Std 141 (Red Book)"	✓
notes[0].text	"For balanced 3\u03c6 loads, Y: V_LL = \u221a3 V_LN, I_L ..."	"For balanced 3\u03c6 loads, Y: V_LL = \u221a3 V_LN, I_L ..."	✓
recommendations.count	1	1	✓
recs[0]	"Memorize: V ratio is \u221a3 in Y; I ratio is \u221a3 in..."	"Memorize: V ratio is \u221a3 in Y; I ratio is \u221a3 in..."	✓
formulas.count	3	3	✓
formulas[0].name	"Y voltage"	"Y voltage"	✓
formulas[0].expression	"V_LL = \u221a3 \u2192 V_LN"	"V_LL = \u221a3 \u2192 V_LN"	✓
formulas[0].variables	"Line vs phase voltage"	"Line vs phase voltage"	✓
formulas[1].name	"\u0394 current"	"\u0394 current"	✓
formulas[1].expression	"I_L = \u221a3 \u2192 I_phase"	"I_L = \u221a3 \u2192 I_phase"	✓
formulas[1].variables	"\u0394: line current is \u221a3 \u2192 phase"	"\u0394: line current is \u221a3 \u2192 phase"	✓
formulas[2].name	"Y current"	"Y current"	✓
formulas[2].expression	"I_L = I_phase"	"I_L = I_phase"	✓
formulas[2].variables	"Y: line = phase current"	"Y: line = phase current"	✓
steps.count	">=0"	0	✓

Balanced Three-Phase Load Calculator — balanced-load

Test: Y · 690V · Z=20Ω · φ=45° · Category: three-phase · Result: **PASS** (39/39 checks)

Input

```
{
  "system": "wye",
  "vll": 690,
  "z": 20,
  "phi": 45
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Phase voltage V_ph".value	398.37	398.37	✓
row:"Phase voltage V_ph".unit	"V"	"V"	✓
row:"Phase current I_ph".value	19.92	19.92	✓
row:"Phase current I_ph".unit	"A"	"A"	✓
row:"Line current I_L".value	19.92	19.92	✓
row:"Line current I_L".unit	"A"	"A"	✓
row:"Total apparent S".value	23.805	23.805	✓
row:"Total apparent S".unit	"kVA"	"kVA"	✓
row:"Total real P".value	16.832	16.833	✓
row:"Total real P".unit	"kW"	"kW"	✓
row:"Total reactive Q".value	16.832	16.833	✓
row:"Total reactive Q".unit	"kVAr"	"kVAr"	✓
status	"ok"	"ok"	✓
summary	"I_L = 19.92 A \u00b7 S = 23.81 kVA \u00b7 PF = 0.707"	"I_L = 19.92 A \u00b7 S = 23.81 kVA \u00b7 PF = 0.707"	✓
notes.count	1	1	✓
notes[0].standard	"IEEE"	"IEEE"	✓
notes[0].reference	"IEEE Std 141"	"IEEE Std 141"	✓
notes[0].text	"For unbalanced loads, use symmetrical components (positi...	"For unbalanced loads, use symmetrical components (positi...	✓
recommendations.count	0	0	✓
formulas.count	6	6	✓
formulas[0].name	"Y: phase voltage"	"Y: phase voltage"	✓
formulas[0].expression	"V_ph = V_LL / \u221a3"	"V_ph = V_LL / \u221a3"	✓
formulas[0].variables	""	""	✓
formulas[1].name	"\u0394: phase voltage"	"\u0394: phase voltage"	✓
formulas[1].expression	"V_ph = V_LL"	"V_ph = V_LL"	✓
formulas[1].variables	""	""	✓

Field	Expected	Actual	Pass
formulas[2].name	"Phase current"	"Phase current"	✓
formulas[2].expression	" $I_{ph} = V_{ph} / Z$ "	" $I_{ph} = V_{ph} / Z$ "	✓
formulas[2].variables	"Z = phase impedance"	"Z = phase impedance"	✓
formulas[3].name	"Y: line current"	"Y: line current"	✓
formulas[3].expression	" $I_L = I_{ph}$ "	" $I_L = I_{ph}$ "	✓
formulas[3].variables	""	""	✓
formulas[4].name	"\u0394: line current"	"\u0394: line current"	✓
formulas[4].expression	" $I_L = \sqrt{3} I_{ph}$ "	" $I_L = \sqrt{3} I_{ph}$ "	✓
formulas[4].variables	""	""	✓
formulas[5].name	"Total power"	"Total power"	✓
formulas[5].expression	" $S = 3 \sqrt{3} V_{ph} I_{ph}$ "	" $S = 3 \sqrt{3} V_{ph} I_{ph}$ "	✓
formulas[5].variables	"For balanced 3\u03c6"	"For balanced 3\u03c6"	✓
steps.count	">=0"	0	✓

Transformer Sizing Calculator — transformer-sizing

Test: 250kW · 480V · 3φ · df 0.9 · grow 15% · Category: transformer · Result: **PASS** (38/38 checks)

Input

```
{
  "P": 250,
  "pf": 0.85,
  "df": 0.9,
  "grow": 15,
  "Vll": 480,
  "eff": 0.97,
  "ph": "3"
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Required apparent S".value	304.41	304.41	✓
row:"Required apparent S".unit	"kVA"	"kVA"	✓
row:"Adjusted for losses".value	313.83	313.83	✓
row:"Adjusted for losses".unit	"kVA"	"kVA"	✓
row:"Standard size".value	500	500	✓
row:"Standard size".unit	"kVA"	"kVA"	✓
row:"Standard size".status	"ok"	"ok"	✓
row:"Loading at full load".value	60.88	60.9	✓
row:"Loading at full load".unit	"%"	"%"	✓
row:"Secondary current".value	601.4	601.4	✓
row:"Secondary current".unit	"A"	"A"	✓
picks.transformerKVA	500	500	✓
status	"ok"	"ok"	✓
summary	"Size 500 kVA (60.9% loaded) \u00b7 601.4 A secondary at ..."	"Size 500 kVA (60.9% loaded) \u00b7 601.4 A secondary at ..."	✓
notes.count	3	3	✓
notes[0].standard	"IEC"	"IEC"	✓
notes[0].reference	"IEC 60076-1"	"IEC 60076-1"	✓
notes[0].text	"Power transformer ratings: standard kVA per IEC 60076 (e..."	"Power transformer ratings: standard kVA per IEC 60076 (e..."	✓
notes[1].standard	"IEEE"	"IEEE"	✓
notes[1].reference	"IEEE C57.12.00"	"IEEE C57.12.00"	✓
notes[1].text	"IEEE standard kVA ratings for liquid-immersed distributi..."	"IEEE standard kVA ratings for liquid-immersed distributi..."	✓
notes[2].standard	"IEC"	"IEC"	✓
notes[2].reference	"IEC 60076-7"	"IEC 60076-7"	✓

Field	Expected	Actual	Pass
notes[2].text	"Loading guide \u2014 nameplate kVA is continuous at rate..."	"Loading guide \u2014 nameplate kVA is continuous at rate..."	✓
recommendations.count	2	2	✓
recs[0]	"Apply 20% spare for future growth unless explicitly acco..."	"Apply 20% spare for future growth unless explicitly acco..."	✓
recs[1]	"For harmonic-rich loads (VFDs, rectifiers), use K-factor..."	"For harmonic-rich loads (VFDs, rectifiers), use K-factor..."	✓
formulas.count	3	3	✓
formulas[0].name	"Apparent load"	"Apparent load"	✓
formulas[0].expression	"S_load = (P/PF) \u00b7 DF \u00b7 (1 + g)"	"S_load = (P/PF) \u00b7 DF \u00b7 (1 + g)"	✓
formulas[0].variables	"g = growth fraction"	"g = growth fraction"	✓
formulas[1].name	"Standard kVA"	"Standard kVA"	✓
formulas[1].expression	"S_tx = next \u2265 S_load / \u03b7"	"S_tx = next \u2265 S_load / \u03b7"	✓
formulas[1].variables	"\u03b7 = transformer efficiency"	"\u03b7 = transformer efficiency"	✓
formulas[2].name	"Loading"	"Loading"	✓
formulas[2].expression	"Loading% = S_load / S_tx"	"Loading% = S_load / S_tx"	✓
formulas[2].variables	""	""	✓
steps.count	">=4"	4	✓

Transformer Primary & Secondary Current — transformer-current

Test: 1000kVA · 22kV/400V · Z=5.75% · 3φ · Category: transformer · Result: **PASS** (38/38 checks)

Input

```
{
  "S": 1000,
  "V1": 22000,
  "V2": 400,
  "Z": 5.75,
  "ph": "3"
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Primary current I■".value	26.24	26.24	✓
row:"Primary current I■".unit	"A"	"A"	✓
row:"Secondary current I■".value	1443.38	1443.38	✓
row:"Secondary current I■".unit	"A"	"A"	✓
row:"Turns ratio (a)".value	55	55	✓
row:"Impedance Z%".value	5.75	5.75	✓
row:"Impedance Z%".unit	"%"	"%"	✓
row:"Short-circuit I_sc".value	25102	25102	✓
row:"Short-circuit I_sc".unit	"A"	"A"	✓
row:"Short-circuit I_sc".status	"warn"	"warn"	✓
row:"Short-circuit MVA".value	17.39	17.39	✓
row:"Short-circuit MVA".unit	"MVA"	"MVA"	✓
status	"ok"	"ok"	✓
summary	"I\u2081 = 26.2 A \u00b7 I\u2082 = 1443.4 A \u00b7 I_sc \u2013"	"I\u2081 = 26.2 A \u00b7 I\u2082 = 1443.4 A \u00b7 I_sc \u2013"	✓
notes.count	2	2	✓
notes[0].standard	"IEEE"	"IEEE"	✓
notes[0].reference	"IEEE C57.12.00"	"IEEE C57.12.00"	✓
notes[0].text	"Distribution transformer impedance typically 5\u20137.5%..."	"Distribution transformer impedance typically 5\u20137.5%..."	✓
notes[1].standard	"IEC"	"IEC"	✓
notes[1].reference	"IEC 60076-5"	"IEC 60076-5"	✓
notes[1].text	"Ability to withstand short-circuit tested per IEC 60076-5."	"Ability to withstand short-circuit tested per IEC 60076-5."	✓
recommendations.count	2	2	✓
recs[0]	"Set LV breaker short-circuit rating \u2265 I_sc (25102 A..."	"Set LV breaker short-circuit rating \u2265 I_sc (25102 A..."	✓

Field	Expected	Actual	Pass
recs[1]	"Verify cable & bus bracing short-circuit withstand \u226...	"Verify cable & bus bracing short-circuit withstand \u226...	✓
formulas.count	4	4	✓
formulas[0].name	"Primary current"	"Primary current"	✓
formulas[0].expression	" $I_{S1} = S / (\sqrt{3} V_{S1})$ "	" $I_{S1} = S / (\sqrt{3} V_{S1})$ "	✓
formulas[0].variables	" S in VA"	" S in VA"	✓
formulas[1].name	"Secondary current"	"Secondary current"	✓
formulas[1].expression	" $I_{S2} = S / (\sqrt{3} V_{S2})$ "	" $I_{S2} = S / (\sqrt{3} V_{S2})$ "	✓
formulas[1].variables	" S "	" S "	✓
formulas[2].name	"Turns ratio"	"Turns ratio"	✓
formulas[2].expression	" $a = V_{S1} / V_{S2}$ "	" $a = V_{S1} / V_{S2}$ "	✓
formulas[2].variables	""	""	✓
formulas[3].name	"Short-circuit"	"Short-circuit"	✓
formulas[3].expression	" $I_{sc} = I_{S2} / Z\%$ "	" $I_{sc} = I_{S2} / Z\%$ "	✓
formulas[3].variables	"Z% on secondary base"	"Z% on secondary base"	✓
steps.count	">=0"	0	✓

Transformer Efficiency & Losses — transformer-efficiency

Test: 500kVA · P0=0.6 · Pk=5.5 · 80% load · Category: transformer · Result: **FAIL** (32/33 checks)

Input

```
{
  "S": 500,
  "P0": 0.6,
  "Pk": 5.5,
  "load": 80,
  "pf": 0.85
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Output P _{out} ".value	340	340	✓
row:"Output P _{out} ".unit	"kW"	"kW"	✓
row:"No-load loss P ₀ ".value	0.6	0.6	✓
row:"No-load loss P ₀ ".unit	"kW"	"kW"	✓
row:"Load loss P _k · L ² ".value	3.52	3.52	✓
row:"Load loss P _k · L ² ".unit	"kW"	"kW"	✓
row:"Total losses".value	4.12	4.12	✓
row:"Total losses".unit	"kW"	"kW"	✓
row:"Efficiency η".value	98.8	98.8	✓
row:"Efficiency η".unit	"%"	"%"	✓
row:"Max-efficiency loading".value	33	33	✓
row:"Max-efficiency loading".unit	"%"	"%"	✓
row:"Efficiency at peak".value	98.8	99.15	✓
row:"Efficiency at peak".unit	"%"	"%"	✓
status	"info"	"ok"	✗
status	"ok"	"ok"	✓
summary	"\u03b7 = 98.8% at 80% load \u00b7 max at 33%"	"\u03b7 = 98.8% at 80% load \u00b7 max at 33%"	✓
notes.count	2	2	✓
notes[0].standard	"IEC"	"IEC"	✓
notes[0].reference	"IEC 60076-1"	"IEC 60076-1"	✓
notes[0].text	"Losses measured per IEC 60076-1. Load loss referenced to..."	"Losses measured per IEC 60076-1. Load loss referenced to..."	✓
notes[1].standard	"IEEE"	"IEEE"	✓
notes[1].reference	"IEEE C57.12.90"	"IEEE C57.12.90"	✓
notes[1].text	"Test code for distribution transformers \u2014 no-load L..."	"Test code for distribution transformers \u2014 no-load L..."	✓

Field	Expected	Actual	Pass
recommendations.count	0	0	✓
formulas.count	2	2	✓
formulas[0].name	"Efficiency"	"Efficiency"	✓
formulas[0].expression	" $\eta = \frac{P_{out}}{(P_{out} + P_{loss} + P_{k})}$ "	" $\eta = \frac{P_{out}}{(P_{out} + P_{loss} + P_{k})}$ "	✓
formulas[0].variables	"L = per-unit load"	"L = per-unit load"	✓
formulas[1].name	"Max efficiency"	"Max efficiency"	✓
formulas[1].expression	" $\eta_{max} = \frac{P_{out}}{P_{k}}$ "	" $\eta_{max} = \frac{P_{out}}{P_{k}}$ "	✓
formulas[1].variables	"where load loss = no-load loss"	"where load loss = no-load loss"	✓
steps.count	">=3"	3	✓

Motor Full-Load Current (FLC) — motor-fl-current

Test: 22kW · 400V · 3 ϕ · pf 0.85 · eff 0.9 (calc) · Category: motor · Result: **PASS** (33/33 checks)

Input

```
{
  "P": 22,
  "V": 400,
  "ph": "3",
  "pf": 0.85,
  "eff": 0.9,
  "standard": "calc"
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Full-load current I_FLC".value	41.51	41.51	✓
row:"Full-load current I_FLC".unit	"A"	"A"	✓
row:"Full-load current I_FLC".status	"ok"	"ok"	✓
row:"Direct calculation".value	41.51	41.51	✓
row:"Direct calculation".unit	"A"	"A"	✓
row:"Recommended OCPD (inverse-time)".value	105	105	✓
row:"Recommended OCPD (inverse-time)".unit	"A"	"A"	✓
row:"Recommended OCPD (inverse-time)".status	"info"	"info"	✓
picks.motorFLC	41.5	41.5	✓
status	"ok"	"ok"	✓
summary	"FLC \u2248 41.5 A \u00b7 OCPD \u2248 105 A (inverse-time)"	"FLC \u2248 41.5 A \u00b7 OCPD \u2248 105 A (inverse-time)"	✓
notes.count	3	3	✓
notes[0].standard	"NEC"	"NEC"	✓
notes[0].reference	"NEC 430.250"	"NEC 430.250"	✓
notes[0].text	"FLC values for Design B, C, and D motors. Use next highe..."	"FLC values for Design B, C, and D motors. Use next highe..."	✓
notes[1].standard	"NEMA"	"NEMA"	✓
notes[1].reference	"NEMA MG-1"	"NEMA MG-1"	✓
notes[1].text	"Defines motor performance including locked-rotor current..."	"Defines motor performance including locked-rotor current..."	✓
notes[2].standard	"IEC"	"IEC"	✓
notes[2].reference	"IEC 60034-1"	"IEC 60034-1"	✓

Field	Expected	Actual	Pass
notes[2].text	"Rotating electrical machines \\u2014 nameplate rating and..."	"Rotating electrical machines \\u2014 nameplate rating and..."	✓
recommendations.count	3	3	✓
recs[0]	"Inverse-time OCPD: next standard size \\u2265 104 A (NEC ..."	"Inverse-time OCPD: next standard size \\u2265 104 A (NEC ..."	✓
recs[1]	"Choose overload relay at 115\\u2013125% of FLC for 1.15 S..."	"Choose overload relay at 115\\u2013125% of FLC for 1.15 S..."	✓
recs[2]	"For VFD-driven motors, set VFD current limit at motor na..."	"For VFD-driven motors, set VFD current limit at motor na..."	✓
formulas.count	2	2	✓
formulas[0].name	"Direct FLC"	"Direct FLC"	✓
formulas[0].expression	"I = P / (\\u221a3 \\u00b7 V_LL \\u00b7 PF \\u00b7 \\u03b7)"	"I = P / (\\u221a3 \\u00b7 V_LL \\u00b7 PF \\u00b7 \\u03b7)"	✓
formulas[0].variables	"3\\u03c6; P in W"	"3\\u03c6; P in W"	✓
formulas[1].name	"NEC OCPD"	"NEC OCPD"	✓
formulas[1].expression	"OCPD \\u2265 2.5 \\u00d7 FLC (Design B)"	"OCPD \\u2265 2.5 \\u00d7 FLC (Design B)"	✓
formulas[1].variables	"NEC 430.52"	"NEC 430.52"	✓
steps.count	">=0"	0	✓

Motor Starting Current & Methods — motor-starting

Test: 45A · LR×7 · DOL · 400V · 30MVA · Category: motor · Result: PASS (37/37 checks)

Input

```
{
  "I": 45,
  "lr": 7,
  "method": "dol",
  "autoPct": 65,
  "V": 400,
  "Ssc": 30
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Locked-rotor current (DOL)".value	315	315	✓
row:"Locked-rotor current (DOL)".unit	"A"	"A"	✓
row:"Locked-rotor current (DOL)".status	"warn"	"warn"	✓
row:"Starting current (method)".value	315	315	✓
row:"Starting current (method)".unit	"A"	"A"	✓
row:"Starting current (method)".status	"ok"	"ok"	✓
row:"Method factor k".value	1	1	✓
row:"Voltage dip on source".value	0.2	0.2	✓
row:"Voltage dip on source".unit	"%"	"%"	✓
row:"Mechanical stress level".value	"High"	"High"	✓
row:"Mechanical stress level".unit	""	""	✓
status	"ok"	"ok"	✓
summary	"I_start = 315 A (DOL) \u00b7 \u0394 \u0394 0.2%"	"I_start = 315 A (DOL) \u00b7 \u0394 \u0394 0.2%"	✓
notes.count	2	2	✓
notes[0].standard	"IEC"	"IEC"	✓
notes[0].reference	"IEC 60034-12"	"IEC 60034-12"	✓
notes[0].text	"Starting performance of cage induction motors \u2014 Des..."	"Starting performance of cage induction motors \u2014 Des..."	✓
notes[1].standard	"NEMA"	"NEMA"	✓
notes[1].reference	"NEMA MG-1 Part 12"	"NEMA MG-1 Part 12"	✓
notes[1].text	"Locked-rotor current codes A\u2013J; Design B typical LR..."	"Locked-rotor current codes A\u2013J; Design B typical LR..."	✓
recommendations.count	2	2	✓

Field	Expected	Actual	Pass
recs[0]	"Star-delta: 6 lead motor required; transitions cause 2nd..."	"Star-delta: 6 lead motor required; transitions cause 2nd..."	✓
recs[1]	"VFD eliminates inrush entirely and provides soft ramp & ..."	"VFD eliminates inrush entirely and provides soft ramp & ..."	✓
formulas.count	4	4	✓
formulas[0].name	"Locked-rotor current"	"Locked-rotor current"	✓
formulas[0].expression	" $I_{LR} = LR \cdot \frac{1}{FLC}$ "	" $I_{LR} = LR \cdot \frac{1}{FLC}$ "	✓
formulas[0].variables	"LR typical 6\20138"	"LR typical 6\20138"	✓
formulas[1].name	"Star-delta start"	"Star-delta start"	✓
formulas[1].expression	" $I_{start} = I_{LR} / 3$ "	" $I_{start} = I_{LR} / 3$ "	✓
formulas[1].variables	""	""	✓
formulas[2].name	"Autotransformer"	"Autotransformer"	✓
formulas[2].expression	" $I_{start} = k \cdot \frac{1}{I_{LR}}$ "	" $I_{start} = k \cdot \frac{1}{I_{LR}}$ "	✓
formulas[2].variables	"k = tap ratio"	"k = tap ratio"	✓
formulas[3].name	"Voltage dip"	"Voltage dip"	✓
formulas[3].expression	" $\frac{U_{394U} \cdot U_{2248}}{I_{start} \cdot I_{sc}}$ "	" $\frac{U_{394U} \cdot U_{2248}}{I_{start} \cdot I_{sc}}$ "	✓
formulas[3].variables	" $I_{sc} = S_{sc} / (\frac{1}{221a3} \cdot \frac{1}{00b7V})$ "	" $I_{sc} = S_{sc} / (\frac{1}{221a3} \cdot \frac{1}{00b7V})$ "	✓
steps.count	">=0"	0	✓

VFD (Variable Frequency Drive) Sizing — vfd-sizing

Test: 45A · 400V · CT · 45°C · 1500m · 6kHz · Category: motor · Result: **PASS** (36/36 checks)

Input

```
{
  "I": 45,
  "V": 400,
  "ph": "3",
  "duty": "CT",
  "Ta": 45,
  "alt": 1500,
  "fc": 6
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Motor FLC".value	45	45	✓
row:"Motor FLC".unit	"A"	"A"	✓
row:"Duty factor (k_d)".value	1.05	1.05	✓
row:"Temp derate (k_T)".value	0.9	0.9	✓
row:"Altitude derate (k_A)".value	0.95	0.95	✓
row:"Carrier derate (k_F)".value	0.95	0.95	✓
row:"Worst derate".value	0.9	0.9	✓
row:"Required VFD current".value	50	50	✓
row:"Required VFD current".unit	"A"	"A"	✓
row:"Required VFD power".value	34.64	34.64	✓
row:"Required VFD power".unit	"kVA"	"kVA"	✓
row:"Required VFD power".status	"ok"	"ok"	✓
status	"ok"	"ok"	✓
summary	"VFD \u2265 50 A (34.64 kVA) at derate 90%"	"VFD \u2265 50 A (34.64 kVA) at derate 90%"	✓
notes.count	2	2	✓
notes[0].standard	"IEC"	"IEC"	✓
notes[0].reference	"IEC 61800-2"	"IEC 61800-2"	✓
notes[0].text	"Adjustable speed electrical power drive systems \u2014 g..."	"Adjustable speed electrical power drive systems \u2014 g..."	✓
notes[1].standard	"NEMA"	"NEMA"	✓
notes[1].reference	"NEMA ICS 7"	"NEMA ICS 7"	✓
notes[1].text	"Industrial control and systems \u2014 adjustable-speed d..."	"Industrial control and systems \u2014 adjustable-speed d..."	✓
recommendations.count	3	3	✓
recs[0]	"Select VFD nominal \u2265 50 A continuous, 75 A peak (60..."	"Select VFD nominal \u2265 50 A continuous, 75 A peak (60..."	✓

Field	Expected	Actual	Pass
recs[1]	"Use inverter-duty cable or output reactor/reactor filter..."	"Use inverter-duty cable or output reactor/reactor filter..."	✓
recs[2]	"Install line reactor or passive harmonic filter at VFD i..."	"Install line reactor or passive harmonic filter at VFD i..."	✓
formulas.count	3	3	✓
formulas[0].name	"Required current"	"Required current"	✓
formulas[0].expression	"I_req = I_FLC / (k_d \u00b7 k_T \u00b7 k_A \u00b7 k_F)"	"I_req = I_FLC / (k_d \u00b7 k_T \u00b7 k_A \u00b7 k_F)"	✓
formulas[0].variables	""	""	✓
formulas[1].name	"Temp derate"	"Temp derate"	✓
formulas[1].expression	"k_T = 1 \u2212 0.02\u00b7(T_a \u2212 40)"	"k_T = 1 \u2212 0.02\u00b7(T_a \u2212 40)"	✓
formulas[1].variables	"\u00b0C above 40"	"\u00b0C above 40"	✓
formulas[2].name	"Altitude derate"	"Altitude derate"	✓
formulas[2].expression	"k_A = 1 \u2212 0.01\u00b7((alt \u2212 1000)/100)"	"k_A = 1 \u2212 0.01\u00b7((alt \u2212 1000)/100)"	✓
formulas[2].variables	"per 100 m above 1000 m"	"per 100 m above 1000 m"	✓
steps.count	">=0"	0	✓

Motor Circuit Breaker / OCPD Sizing — motor-breaker

Test: 45A · inverse-time · 3P · Category: motor · Result: **PASS** (42/42 checks)

Input

```
{
  "I": 45,
  "type": "inv",
  "poles": "3"
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Motor FLC".value	45	45	✓
row:"Motor FLC".unit	"A"	"A"	✓
row:"Protection factor (k)".value	2.5	2.5	✓
row:"Target OCPD".value	112.5	112.5	✓
row:"Target OCPD".unit	"A"	"A"	✓
row:"Selected OCPD (next std)".value	125	125	✓
row:"Selected OCPD (next std)".unit	"A"	"A"	✓
row:"Selected OCPD (next std)".status	"ok"	"ok"	✓
row:"Overload relay setting".value	"51.7 \u2013 56.3"	"51.7 \u2013 56.3"	✓
row:"Overload relay setting".unit	"A"	"A"	✓
row:"Overload relay setting".status	"info"	"info"	✓
row:"Min conductor ampacity".value	56.3	56.3	✓
row:"Min conductor ampacity".unit	"A"	"A"	✓
picks.breakerRating	125	125	✓
picks.breakerFrame	125	125	✓
status	"ok"	"ok"	✓
summary	"OCPD 125 A \u00b7 OL 51.7\u201356.3 A \u00b7 cable \u2264"	"OCPD 125 A \u00b7 OL 51.7\u201356.3 A \u00b7 cable \u2264"	✓
notes.count	2	2	✓
notes[0].standard	"NEC"	"NEC"	✓
notes[0].reference	"NEC 430.52"	"NEC 430.52"	✓
notes[0].text	"Branch-circuit short-circuit & ground-fault protection: ..."	"Branch-circuit short-circuit & ground-fault protection: ..."	✓
notes[1].standard	"IEC"	"IEC"	✓
notes[1].reference	"IEC 60947-4-1"	"IEC 60947-4-1"	✓

Field	Expected	Actual	Pass
notes[1].text	"Type 1 / Type 2 coordination. Type 2 = no welding, trip ..."	"Type 1 / Type 2 coordination. Type 2 = no welding, trip ..."	✓
recommendations.count	3	3	✓
recs[0]	"Breaker: 125 A, 3-pole."	"Breaker: 125 A, 3-pole."	✓
recs[1]	"Set overload to motor nameplate FLA (1.0 SF motor) or 11..."	"Set overload to motor nameplate FLA (1.0 SF motor) or 11..."	✓
recs[2]	"Verify SC rating \u2265 available fault current at line ..."	"Verify SC rating \u2265 available fault current at line ..."	✓
formulas.count	4	4	✓
formulas[0].name	"OCPD (inverse-time)"	"OCPD (inverse-time)"	✓
formulas[0].expression	"OCPD \u2265 2.5 \u00d7 FLC"	"OCPD \u2265 2.5 \u00d7 FLC"	✓
formulas[0].variables	"NEC 430.52, Design B"	"NEC 430.52, Design B"	✓
formulas[1].name	"OCPD (instantaneous)"	"OCPD (instantaneous)"	✓
formulas[1].expression	"OCPD \u2265 8 \u00d7 FLC"	"OCPD \u2265 8 \u00d7 FLC"	✓
formulas[1].variables	"Motor circuit protector"	"Motor circuit protector"	✓
formulas[2].name	"OCPD (dual fuse)"	"OCPD (dual fuse)"	✓
formulas[2].expression	"OCPD \u2265 1.75 \u00d7 FLC"	"OCPD \u2265 1.75 \u00d7 FLC"	✓
formulas[2].variables	"Time-delay fuse"	"Time-delay fuse"	✓
formulas[3].name	"Conductor"	"Conductor"	✓
formulas[3].expression	"A \u2265 1.25 \u00d7 FLC"	"A \u2265 1.25 \u00d7 FLC"	✓
formulas[3].variables	"Continuous load"	"Continuous load"	✓
steps.count	">=0"	0	✓

Voltage Drop Calculator — voltage-drop

Test: 400V · 3φ · 80A · 60m · 16mm² Cu · 60°C · Category: cable · Result: **PASS** (39/39 checks)

Input

```
{
  "V": 400,
  "ph": "3",
  "I": 80,
  "L": 60,
  "mat": "Cu",
  "size": 16,
  "pf": 0.9,
  "T": 60
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Voltage drop ΔU".value	9.62	9.62	✓
row:"Voltage drop ΔU".unit	"V"	"V"	✓
row:"Voltage drop".value	2.41	2.41	✓
row:"Voltage drop".unit	"%"	"%"	✓
row:"Resistance R_ph".value	0.075	0.075	✓
row:"Resistance R_ph".unit	"\u003a9"	"\u003a9"	✓
row:"Reactance X_ph".value	0.005	0.005	✓
row:"Reactance X_ph".unit	"\u003a9"	"\u003a9"	✓
row:"Voltage at load".value	390.38	390.4	✓
row:"Voltage at load".unit	"V"	"V"	✓
row:"Recommended size for ≤ 3% Vd".value	16	16	✓
row:"Recommended size for ≤ 3% Vd".unit	"mm\u00b2"	"mm\u00b2"	✓
row:"Recommended OCPD (125% I)".value	100	100	✓
row:"Recommended OCPD (125% I)".unit	"A"	"A"	✓
picks.cableSize	16	16	✓
picks.breakerRating	100	100	✓
status	"ok"	"ok"	✓
summary	"\u00394U = 2.41% (9.6 V) \u00b7 16 mm\u00b2 meets \u2264 ..."	"\u00394U = 2.41% (9.6 V) \u00b7 16 mm\u00b2 meets \u2264 ..."	✓
notes.count	2	2	✓
notes[0].standard	"NEC"	"NEC"	✓
notes[0].reference	"NEC 210.19 / 215.2"	"NEC 210.19 / 215.2"	✓
notes[0].text	"Branch circuit / feeder Vd \u2264 3% for furthest outlet..."	"Branch circuit / feeder Vd \u2264 3% for furthest outlet..."	✓

Field	Expected	Actual	Pass
notes[1].standard	"IEC"	"IEC"	✓
notes[1].reference	"IEC 60364-5-52"	"IEC 60364-5-52"	✓
notes[1].text	"Annex G provides Vd calculation method including tempera..."	"Annex G provides Vd calculation method including tempera..."	✓
recommendations.count	2	2	✓
recs[0]	"Vd% \u2264 3% \u2014 within typical branch-circuit target."	"Vd% \u2264 3% \u2014 within typical branch-circuit target."	✓
recs[1]	"Verify cable ampacity separately (use cable-sizing calcu..."	"Verify cable ampacity separately (use cable-sizing calcu..."	✓
formulas.count	3	3	✓
formulas[0].name	"Voltage drop"	"Voltage drop"	✓
formulas[0].expression	"\u0394U = k\u00b7I \u00b7(R\u00b7cos \u03c6 + X\u00b7sin ..."	"\u0394U = k\u00b7I \u00b7(R\u00b7cos \u03c6 + X\u00b7sin ..."	✓
formulas[0].variables	"k = 2 (1\u03c6), \u221a3 (3\u03c6)"	"k = 2 (1\u03c6), \u221a3 (3\u03c6)"	✓
formulas[1].name	"Operating R"	"Operating R"	✓
formulas[1].expression	"R_T = R\u2082\u2080\u2080\u00b7(1 + \u03b1\u00b7(T \u2212 20))"	"R_T = R\u2082\u2080\u2080\u00b7(1 + \u03b1\u00b7(T \u2212 20))"	✓
formulas[1].variables	"\u03b1 = 0.00393 (Cu) / 0.00403 (Al)"	"\u03b1 = 0.00393 (Cu) / 0.00403 (Al)"	✓
formulas[2].name	"Vd %"	"Vd %"	✓
formulas[2].expression	"Vd% = \u0394U / V_nom \u00b7 100"	"Vd% = \u0394U / V_nom \u00b7 100"	✓
formulas[2].variables	"3% branch / 5% total"	"3% branch / 5% total"	✓
steps.count	">=4"	4	✓

Cable Sizing Calculator — cable-sizing

Test: 100A · XLPE · C · Cu · 35°C · 2 circuits · Category: cable · Result: **PASS** (47/47 checks)

Input

```
{
  "IB": 100,
  "ins": "XLPE",
  "method": "C",
  "mat": "Cu",
  "Ta": 35,
  "grp": 2,
  "soil": 1.2,
  "duty": "cont"
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Design current I_B".value	100	100	✓
row:"Design current I_B".unit	"A"	"A"	✓
row:"Continuous-load factor k _■ ".value	1.25	1.25	✓
row:"Required I_n".value	125	125	✓
row:"Required I_n".unit	"A"	"A"	✓
row:"Temperature factor k _T ".value	0.96	0.96	✓
row:"Grouping factor k _G ".value	0.88	0.88	✓
row:"Soil derate k _S ".value	1	1	✓
row:"Required I_z (ampacity)".value	147.96	148	✓
row:"Required I_z (ampacity)".unit	"A"	"A"	✓
row:"Selected size".value	35	35	✓
row:"Selected size".unit	"mm\u00b2"	"mm\u00b2"	✓
row:"Selected size".status	"ok"	"ok"	✓
row:"Cable ampacity (I_z)".value	168	168	✓
row:"Cable ampacity (I_z)".unit	"A"	"A"	✓
row:"Cable ampacity (I_z)".status	"ok"	"ok"	✓
row:"Recommended OCPD (next std)".value	125	125	✓
row:"Recommended OCPD (next std)".unit	"A"	"A"	✓
picks.cableSize	35	35	✓
picks.breakerRating	125	125	✓
status	"ok"	"ok"	✓
summary	"Select 35 mm\u00b2 Cu (XLPE, method C) \u00b7 OCPD 125 A"	"Select 35 mm\u00b2 Cu (XLPE, method C) \u00b7 OCPD 125 A"	✓
notes.count	2	2	✓
notes[0].standard	"IEC"	"IEC"	✓

Field	Expected	Actual	Pass
notes[0].reference	"IEC 60364-5-52"	"IEC 60364-5-52"	✓
notes[0].text	"Tables B.52.2 to B.52.14 give ampacity for installation ..."	"Tables B.52.2 to B.52.14 give ampacity for installation ..."	✓
notes[1].standard	"NEC"	"NEC"	✓
notes[1].reference	"NEC 310.15(B)(16)"	"NEC 310.15(B)(16)"	✓
notes[1].text	"Allowable ampacity for 60/75/90 \u00b0C conductors. Appl..."	"Allowable ampacity for 60/75/90 \u00b0C conductors. Appl..."	✓
recommendations.count	3	3	✓
recs[0]	"Voltage drop must be checked separately (use Voltage Dro..."	"Voltage drop must be checked separately (use Voltage Dro..."	✓
recs[1]	"Verify short-circuit withstand: I\u00b2t cable \u2265 I\u2265 I\u2265..."	"Verify short-circuit withstand: I\u00b2t cable \u2265 I\u2265 I\u2265..."	✓
recs[2]	"For long parallel runs, consider derating per ambient/bu..."	"For long parallel runs, consider derating per ambient/bu..."	✓
formulas.count	4	4	✓
formulas[0].name	"Required ampacity"	"Required ampacity"	✓
formulas[0].expression	"I_z \u2265 k\u2081 \u00b7 I_B / (k_T \u00b7 k_G \u00b7 k..."	"I_z \u2265 k\u2081 \u00b7 I_B / (k_T \u00b7 k_G \u00b7 k..."	✓
formulas[0].variables	"IEC 60364-5-52"	"IEC 60364-5-52"	✓
formulas[1].name	"Temperature"	"Temperature"	✓
formulas[1].expression	"k_T (B.52.14)"	"k_T (B.52.14)"	✓
formulas[1].variables	"From insulation & ambient"	"From insulation & ambient"	✓
formulas[2].name	"Grouping"	"Grouping"	✓
formulas[2].expression	"k_G (B.52.17)"	"k_G (B.52.17)"	✓
formulas[2].variables	"Number of circuits"	"Number of circuits"	✓
formulas[3].name	"Continuous load"	"Continuous load"	✓
formulas[3].expression	"k\u2081 = 1.25"	"k\u2081 = 1.25"	✓
formulas[3].variables	"NEC 210.20 / 215.3"	"NEC 210.20 / 215.3"	✓
steps.count	">=0"	0	✓

Cable Derating Factors Calculator — cable-derating

Test: 35mm² PVC · C · 45°C · 3 circuits · soil 1.5 · Category: cable · Result: **PASS** (24/24 checks)

Input

```
{
  "size": 35,
  "ins": "PVC",
  "method": "C",
  "Ta": 45,
  "grp": 3,
  "soil": 1.5
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Temperature factor k_T".value	0.79	0.79	✓
row:"Grouping factor k_G".value	0.82	0.82	✓
row:"Soil factor k_S".value	0.93	0.93	✓
row:"Combined derate K".value	0.602	0.602	✓
row:"Base ampacity I_z,table".value	148	148	✓
row:"Base ampacity I_z,table".unit	"A"	"A"	✓
row:"Final ampacity I_z'".value	89.1	89.2	✓
row:"Final ampacity I_z'".unit	"A"	"A"	✓
row:"Final ampacity I_z'".status	"ok"	"ok"	✓
status	"warn"	"warn"	✓
summary	"K = 0.602 \u00b7 I_z' = 89.2 A (from 148 A table)"	"K = 0.602 \u00b7 I_z' = 89.2 A (from 148 A table)"	✓
notes.count	1	1	✓
notes[0].standard	"IEC"	"IEC"	✓
notes[0].reference	"IEC 60364-5-52 \u00a7523"	"IEC 60364-5-52 \u00a7523"	✓
notes[0].text	"Apply derating factors where installation conditions dif..."	"Apply derating factors where installation conditions dif..."	✓
recommendations.count	0	0	✓
formulas.count	2	2	✓
formulas[0].name	"Combined derate"	"Combined derate"	✓
formulas[0].expression	"K = k_T \u00b7 k_G \u00b7 k_S"	"K = k_T \u00b7 k_G \u00b7 k_S"	✓
formulas[0].variables	""	""	✓
formulas[1].name	"Final ampacity"	"Final ampacity"	✓
formulas[1].expression	"I_z' = I_z,table \u00b7 K"	"I_z' = I_z,table \u00b7 K"	✓
formulas[1].variables	""	""	✓
steps.count	">=0"	0	✓

Short-Circuit Current (I²t) Withstand — short-circuit

Test: 25kA · 35mm² Cu/XLPE · 0.3s · device I²t=200M (FAIL) · Category: cable · Result: **PASS** (32/32 checks)

Input

```
{
  "Isc": 25000,
  "S": 35,
  "mat": "Cu",
  "ins": "XLPE",
  "t": 0.3,
  "I2t_dev": 200000000
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Material constant k".value	143	143	✓
row:"Cable I²t withstand (k²·S²)".value	25.05	25.05	✓
row:"Cable I²t withstand (k²·S²)".unit	"kA\u00b2s"	"kA\u00b2s"	✓
row:"Cable I²t withstand (k²·S²)".status	"ok"	"ok"	✓
row:"Source I²t".value	187.5	187.5	✓
row:"Source I²t".unit	"kA\u00b2s"	"kA\u00b2s"	✓
row:"Device I²t (given)".value	200	200	✓
row:"Device I²t (given)".unit	"kA\u00b2s"	"kA\u00b2s"	✓
row:"Verdict".value	"FAIL \u00b7 cable will be damaged"	"FAIL \u00b7 cable will be damaged"	✓
row:"Verdict".unit	""	""	✓
row:"Verdict".status	"err"	"err"	✓
status	"err"	"err"	✓
summary	"Cable I\u00b2t = 25.05 kA\u00b2s vs 200 kA\u00b2s device"	"Cable I\u00b2t = 25.05 kA\u00b2s vs 200 kA\u00b2s device"	✓
notes.count	2	2	✓
notes[0].standard	"IEC"	"IEC"	✓
notes[0].reference	"IEC 60364-4-43 \u00a7433"	"IEC 60364-4-43 \u00a7433"	✓
notes[0].text	"Adiabatic equation: S \u2265 I\u00b7\u221at / k. k depen..."	"Adiabatic equation: S \u2265 I\u00b7\u221at / k. k depen..."	✓
notes[1].standard	"NEC"	"NEC"	✓
notes[1].reference	"NEC 110.10"	"NEC 110.10"	✓
notes[1].text	"Circuit impedance and other characteristics shall be sel..."	"Circuit impedance and other characteristics shall be sel..."	✓
recommendations.count	3	3	✓

Field	Expected	Actual	Pass
recs[0]	"Cable undersized for SC \u2014 increase cross-section or..."	"Cable undersized for SC \u2014 increase cross-section or..."	✓
recs[1]	"Always cross-check with the actual protective device let..."	"Always cross-check with the actual protective device let..."	✓
recs[2]	"For long cable runs, source impedance reduces I_sc at th..."	"For long cable runs, source impedance reduces I_sc at th..."	✓
formulas.count	2	2	✓
formulas[0].name	"Adiabatic withstand"	"Adiabatic withstand"	✓
formulas[0].expression	"I\u00b2t \u2264 k\u00b2 \u00b7 S\u00b2"	"I\u00b2t \u2264 k\u00b2 \u00b7 S\u00b2"	✓
formulas[0].variables	"k = 115 (PVC/Cu), 143 (XLPE/Cu)"	"k = 115 (PVC/Cu), 143 (XLPE/Cu)"	✓
formulas[1].name	"Source I\u00b2t"	"Source I\u00b2t"	✓
formulas[1].expression	"I\u00b2t = I_sc\u00b2 \u00b7 t"	"I\u00b2t = I_sc\u00b2 \u00b7 t"	✓
formulas[1].variables	"t = clearing time"	"t = clearing time"	✓
steps.count	">=0"	0	✓

Circuit Breaker Sizing — breaker-sizing

Test: 120A cont + 30A non · LV · Category: protection · Result: **PASS** (33/33 checks)

Input

```
{
  "Icont": 120,
  "Inon": 30,
  "system": "LV"
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Continuous load".value	120	120	✓
row:"Continuous load".unit	"A"	"A"	✓
row:"Non-continuous load".value	30	30	✓
row:"Non-continuous load".unit	"A"	"A"	✓
row:"Total (NEC 80% rule)".value	180	180	✓
row:"Total (NEC 80% rule)".unit	"A"	"A"	✓
row:"OCPD (next std)".value	200	200	✓
row:"OCPD (next std)".unit	"A"	"A"	✓
row:"OCPD (next std)".status	"ok"	"ok"	✓
row:"Loading".value	90	90	✓
row:"Loading".unit	"%"	"%"	✓
picks.breakerRating	200	200	✓
picks.breakerFrame	200	200	✓
status	"ok"	"ok"	✓
summary	"Breaker 200 A \u00b7 90% loaded"	"Breaker 200 A \u00b7 90% loaded"	✓
notes.count	2	2	✓
notes[0].standard	"NEC"	"NEC"	✓
notes[0].reference	"NEC 210.20"	"NEC 210.20"	✓
notes[0].text	"OCPD rating \u2265 noncontinuous + 125% continuous load."	"OCPD rating \u2265 noncontinuous + 125% continuous load."	✓
notes[1].standard	"IEC"	"IEC"	✓
notes[1].reference	"IEC 60898 / 60947-2"	"IEC 60898 / 60947-2"	✓
notes[1].text	"Circuit-breakers for overcurrent protection. Tripping ch...	"Circuit-breakers for overcurrent protection. Tripping ch...	✓
recommendations.count	2	2	✓
recs[0]	"Verify SC rating \u2265 available fault current at point..."	"Verify SC rating \u2265 available fault current at point..."	✓

Field	Expected	Actual	Pass
recs[1]	"For motor loads, use inverse-time OCPD per NEC 430.52 (t...	"For motor loads, use inverse-time OCPD per NEC 430.52 (t...	✓
formulas.count	2	2	✓
formulas[0].name	"OCPD"	"OCPD"	✓
formulas[0].expression	"I_OCPD \u2265 1.25\u00b7Icont + Inon"	"I_OCPD \u2265 1.25\u00b7Icont + Inon"	✓
formulas[0].variables	"NEC 210.20 / 215.3"	"NEC 210.20 / 215.3"	✓
formulas[1].name	"Next standard"	"Next standard"	✓
formulas[1].expression	"next \u2265 target"	"next \u2265 target"	✓
formulas[1].variables	"Standard sizes"	"Standard sizes"	✓
steps.count	">=0"	0	✓

Fuse Sizing Calculator — fuse-sizing

Test: 40A cont + 15A non · fast-acting · Category: protection · Result: **PASS** (34/34 checks)

Input

```
{
  "Icont": 40,
  "Inon": 15,
  "type": "FA",
  "inrush": 5
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Continuous load".value	40	40	✓
row:"Continuous load".unit	"A"	"A"	✓
row:"Non-continuous".value	15	15	✓
row:"Non-continuous".unit	"A"	"A"	✓
row:"Multiplier (1.25 / 1.15)".value	1.25	1.25	✓
row:"Computed minimum".value	65	65	✓
row:"Computed minimum".unit	"A"	"A"	✓
row:"Recommended fuse".value	70	70	✓
row:"Recommended fuse".unit	"A"	"A"	✓
row:"Recommended fuse".status	"ok"	"ok"	✓
status	"ok"	"ok"	✓
summary	"Recommend 70 A fast-acting"	"Recommend 70 A fast-acting"	✓
notes.count	2	2	✓
notes[0].standard	"IEC"	"IEC"	✓
notes[0].reference	"IEC 60269-1/2"	"IEC 60269-1/2"	✓
notes[0].text	"Low-voltage fuses \u2014 gG (general purpose), aM (motor..."	"Low-voltage fuses \u2014 gG (general purpose), aM (motor..."	✓
notes[1].standard	"NEC"	"NEC"	✓
notes[1].reference	"NEC 240.4(B)"	"NEC 240.4(B)"	✓
notes[1].text	"Conductor ampacity vs fuse rating \u2014 next higher sta..."	"Conductor ampacity vs fuse rating \u2014 next higher sta..."	✓
recommendations.count	3	3	✓
recs[0]	"Select fuse class to match available fuseholders (CC, J, ..."	"Select fuse class to match available fuseholders (CC, J, ..."	✓
recs[1]	"For DC circuits, select DC-rated fuse (different arcing ..."	"For DC circuits, select DC-rated fuse (different arcing ..."	✓

Field	Expected	Actual	Pass
recs[2]	"Time-delay fuses (dual-element) tolerate motor inrush; f...	"Time-delay fuses (dual-element) tolerate motor inrush; f...	✓
formulas.count	3	3	✓
formulas[0].name	"Fast-acting"	"Fast-acting"	✓
formulas[0].expression	"I_F \u2265 1.25\u00b7I _{cont} + I _{non} "	"I_F \u2265 1.25\u00b7I _{cont} + I _{non} "	✓
formulas[0].variables	"NEC 240.4(B) / 430.52"	"NEC 240.4(B) / 430.52"	✓
formulas[1].name	"Time-delay"	"Time-delay"	✓
formulas[1].expression	"I_F \u2265 1.15\u00b7I _{cont} + I _{non} "	"I_F \u2265 1.15\u00b7I _{cont} + I _{non} "	✓
formulas[1].variables	"Allows harmless inrush"	"Allows harmless inrush"	✓
formulas[2].name	"Motor (aM)"	"Motor (aM)"	✓
formulas[2].expression	"I_F \u2265 I _{cont} + inrush, sized for locked rotor"	"I_F \u2265 I _{cont} + inrush, sized for locked rotor"	✓
formulas[2].variables	"IEC 60269 aM"	"IEC 60269 aM"	✓
steps.count	">=0"	0	✓

Earthing Conductor Sizing — earthing-conductor

Test: 6000A · 0.3s · Cu/XLPE · Category: grounding · Result: **PASS** (28/28 checks)

Input

```
{
  "Isc": 6000,
  "t": 0.3,
  "mat": "Cu",
  "ins": "XLPE"
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Earth-fault current I".value	6000	6000	✓
row:"Earth-fault current I".unit	"A"	"A"	✓
row:"Disconnect time t".value	0.3	0.3	✓
row:"Disconnect time t".unit	"s"	"s"	✓
row:"k coefficient".value	143	143	✓
row:"Computed S (adiabatic)".value	22.98	22.98	✓
row:"Computed S (adiabatic)".unit	"mm\u00b2"	"mm\u00b2"	✓
row:"Next standard size".value	25	25	✓
row:"Next standard size".unit	"mm\u00b2"	"mm\u00b2"	✓
row:"Next standard size".status	"ok"	"ok"	✓
picks.cableSize	25	25	✓
status	"ok"	"ok"	✓
summary	"PE conductor \u2265 25 mm\u00b2 Cu"	"PE conductor \u2265 25 mm\u00b2 Cu"	✓
notes.count	2	2	✓
notes[0].standard	"IEC"	"IEC"	✓
notes[0].reference	"IEC 60364-5-54 \u00a7543.1"	"IEC 60364-5-54 \u00a7543.1"	✓
notes[0].text	"PE conductor must withstand adiabatic I\u00b2t for the d..."	"PE conductor must withstand adiabatic I\u00b2t for the d..."	✓
notes[1].standard	"IEEE"	"IEEE"	✓
notes[1].reference	"IEEE 80 (substation) "	"IEEE 80 (substation) "	✓
notes[1].text	"Step & touch potentials for substations \u2014 separate ..."	"Step & touch potentials for substations \u2014 separate ..."	✓
recommendations.count	2	2	✓
recs[0]	"Verify PE conductor is copper or aluminum \u2014 steel e..."	"Verify PE conductor is copper or aluminum \u2014 steel e..."	✓

Field	Expected	Actual	Pass
recs[1]	"For TN systems, PE sized equal to phase conductor up to ..."	"For TN systems, PE sized equal to phase conductor up to ..."	✓
formulas.count	1	1	✓
formulas[0].name	"Adiabatic S"	"Adiabatic S"	✓
formulas[0].expression	"S = I \u00b7 \u221at / k"	"S = I \u00b7 \u221at / k"	✓
formulas[0].variables	"k = 115 (Cu/PVC) or 143 (Cu/XLPE)"	"k = 115 (Cu/PVC) or 143 (Cu/XLPE)"	✓
steps.count	">=0"	0	✓

Ground Electrode Resistance Estimator — ground-resistance

Test: Single rod · $\rho=50$ · $L=3\text{m}$ · Category: grounding · Result: **PASS** (32/32 checks)

Input

```
{
  "type": "rod",
  "rho": 50,
  "L": 3,
  "d": 0.02,
  "W": 0.6,
  "A": 1,
  "n": 1,
  "s": 3
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Single electrode R".value	16.97	16.97	✓
row:"Single electrode R".unit	"\u03a9"	"\u03a9"	✓
row:"Multi-rod R".value	16.97	16.97	✓
row:"Multi-rod R".unit	"\u03a9"	"\u03a9"	✓
row:"Soil ρ ".value	50	50	✓
row:"Soil ρ ".unit	"\u03a9\u00b7m"	"\u03a9\u00b7m"	✓
row:"Length L".value	3	3	✓
row:"Length L".unit	"m"	"m"	✓
status	"info"	"info"	✓
summary	"R \u2248 17 \u03a9 with 1 rod(s)"	"R \u2248 17 \u03a9 with 1 rod(s)"	✓
notes.count	2	2	✓
notes[0].standard	"IEEE"	"IEEE"	✓
notes[0].reference	"IEEE 80 \u00a711"	"IEEE 80 \u00a711"	✓
notes[0].text	"Detailed formulas for substation ground grid design. Inc..."	"Detailed formulas for substation ground grid design. Inc..."	✓
notes[1].standard	"IEC"	"IEC"	✓
notes[1].reference	"IEC 62561-1"	"IEC 62561-1"	✓
notes[1].text	"Lightning protection system components \u2014 earth elec..."	"Lightning protection system components \u2014 earth elec..."	✓
recommendations.count	3	3	✓
recs[0]	"R > 10 \u03a9 \u2014 typical limit for commercial/indust..."	"R > 10 \u03a9 \u2014 typical limit for commercial/indust..."	✓
recs[1]	"Always measure R after installation \u2014 calculated va..."	"Always measure R after installation \u2014 calculated va..."	✓

Field	Expected	Actual	Pass
recs[2]	"Seasonal variation can be 2\u20134 design f..."	"Seasonal variation can be 2\u20134 design f..."	✓
formulas.count	3	3	✓
formulas[0].name	"Vertical rod"	"Vertical rod"	✓
formulas[0].expression	"R = \u03c1/(2\u03c0L) \u00b7 ln(4L/d)"	"R = \u03c1/(2\u03c0L) \u00b7 ln(4L/d)"	✓
formulas[0].variables	""	""	✓
formulas[1].name	"Multiple rods"	"Multiple rods"	✓
formulas[1].expression	"R_n \u2248 R / n \u00b7 (1 + \u03c1/ (\u03c0\u00b7s\u00b7 7..."	"R_n \u2248 R / n \u00b7 (1 + \u03c1/ (\u03c0\u00b7s\u00b7 7..."	✓
formulas[1].variables	"Mutual coupling correction"	"Mutual coupling correction"	✓
formulas[2].name	"Horizontal bed"	"Horizontal bed"	✓
formulas[2].expression	"Schwarz formula"	"Schwarz formula"	✓
formulas[2].variables	"See IEEE 80"	"See IEEE 80"	✓
steps.count	">=0"	0	✓

Solar Inverter Sizing — solar-inverter

Test: 12kWp · DC/AC 1.15 · Category: renewable · Result: **PASS** (28/28 checks)

Input

```
{
  "Pdc": 12,
  "ratio": 1.15,
  "Vmp": 200,
  "Voc": 600,
  "ph": "1"
}
```

Validation checks

Field	Expected	Actual	Pass
row:"PV DC power".value	12	12	✓
row:"PV DC power".unit	"kWp"	"kWp"	✓
row:"DC/AC ratio".value	1.15	1.15	✓
row:"Inverter AC".value	10.43	10.43	✓
row:"Inverter AC".unit	"kW"	"kW"	✓
row:"Inverter AC".status	"ok"	"ok"	✓
row:"MPPT V range".value	200	200	✓
row:"MPPT V range".unit	"V"	"V"	✓
row:"Max Voc".value	600	600	✓
row:"Max Voc".unit	"V"	"V"	✓
status	"ok"	"ok"	✓
summary	"Inverter \u2265 10.43 kW (DC/AC = 1.15)"	"Inverter \u2265 10.43 kW (DC/AC = 1.15)"	✓
notes.count	2	2	✓
notes[0].standard	"IEEE"	"IEEE"	✓
notes[0].reference	"IEEE 1547-2018"	"IEEE 1547-2018"	✓
notes[0].text	"Standard for interconnecting distributed resources with ..."	"Standard for interconnecting distributed resources with ..."	✓
notes[1].standard	"IEC"	"IEC"	✓
notes[1].reference	"IEC 62109-1/2"	"IEC 62109-1/2"	✓
notes[1].text	"Safety of power converters for use in photovoltaic power..."	"Safety of power converters for use in photovoltaic power..."	✓
recommendations.count	3	3	✓
recs[0]	"Confirm inverter MPPT range covers string Vmp at min exp..."	"Confirm inverter MPPT range covers string Vmp at min exp..."	✓
recs[1]	"Confirm Voc (cold) \u2264 inverter max DC input voltage ..."	"Confirm Voc (cold) \u2264 inverter max DC input voltage ..."	✓

Field	Expected	Actual	Pass
recs[2]	"In high-irradiance regions, DC/AC > 1.2 is common to rec..."	"In high-irradiance regions, DC/AC > 1.2 is common to rec..."	✓
formulas.count	1	1	✓
formulas[0].name	"Inverter AC"	"Inverter AC"	✓
formulas[0].expression	"P_ac = P_dc / (DC/AC ratio)"	"P_ac = P_dc / (DC/AC ratio)"	✓
formulas[0].variables	""	""	✓
steps.count	">=0"	0	✓

Battery Runtime Calculator — battery-runtime

Test: 10kWh · 1.5kW · 80% DoD · 90% eff · Category: renewable · Result: **PASS** (28/28 checks)

Input

```
{
  "E": 10,
  "dod": 80,
  "P": 1.5,
  "eta": 90,
  "age": 0
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Usable energy".value	8	8	✓
row:"Usable energy".unit	"kWh"	"kWh"	✓
row:"Effective load".value	1.667	1.667	✓
row:"Effective load".unit	"kW"	"kW"	✓
row:"Runtime".value	4.8	4.8	✓
row:"Runtime".unit	"h"	"h"	✓
row:"Runtime".status	"ok"	"ok"	✓
status	"ok"	"ok"	✓
summary	"Runtime \u2248 4.8 h at 1.5 kW load"	"Runtime \u2248 4.8 h at 1.5 kW load"	✓
notes.count	2	2	✓
notes[0].standard	"IEEE"	"IEEE"	✓
notes[0].reference	"IEEE 1188-2005"	"IEEE 1188-2005"	✓
notes[0].text	"Lead-acid battery sizing for standby applications. Allow..."	"Lead-acid battery sizing for standby applications. Allow..."	✓
notes[1].standard	"IEC"	"IEC"	✓
notes[1].reference	"IEC 61427"	"IEC 61427"	✓
notes[1].text	"Secondary cells for PV energy systems."	"Secondary cells for PV energy systems."	✓
recommendations.count	3	3	✓
recs[0]	"Apply aging derate at end-of-life (typically 70\u2013138%..."	"Apply aging derate at end-of-life (typically 70\u2013138%..."	✓
recs[1]	"For UPS, design for \u2265 15 min runtime at full load."	"For UPS, design for \u2265 15 min runtime at full load."	✓
recs[2]	"LiFeP04 supports deeper DoD (90%) vs lead-acid (50%)."	"LiFeP04 supports deeper DoD (90%) vs lead-acid (50%)."	✓
formulas.count	2	2	✓
formulas[0].name	"Usable energy"	"Usable energy"	✓

Field	Expected	Actual	Pass
formulas[0].expression	"E_usable = E_nom \u00b7 DoD \u00b7 (1 \u2212 aging)"	"E_usable = E_nom \u00b7 DoD \u00b7 (1 \u2212 aging)"	✓
formulas[0].variables	""	""	✓
formulas[1].name	"Runtime"	"Runtime"	✓
formulas[1].expression	"t = E_usable / (P_load / \u03b7)"	"t = E_usable / (P_load / \u03b7)"	✓
formulas[1].variables	""	""	✓
steps.count	">=0"	0	✓

Solar String Sizing Calculator — solar-string

Test: Vmp=41 · Voc=49 · -10°C to 70°C · Category: renewable · Result: **PASS** (36/36 checks)

Input

```
{
  "Vmp_mod": 41,
  "Voc_mod": 49,
  "Tc_min": -10,
  "Tc_max": 70,
  "Vmppt_min": 200,
  "Vmppt_max": 800,
  "Vinv_max": 1000
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Vmp at hot cell (Tmax)".value	34.54	34.54	✓
row:"Vmp at hot cell (Tmax)".unit	"V"	"V"	✓
row:"Voc at cold cell (Tmin)".value	55	55	✓
row:"Voc at cold cell (Tmin)".unit	"V"	"V"	✓
row:"Min modules (hot Vmp ≥ MPPT min)".value	6	6	✓
row:"Max modules (cold Voc ≤ inverter max)".value	18	18	✓
row:"Max modules (Vmp ≤ MPPT max)".value	19	19	✓
row:"Recommended range".value	"6 \u2013 18"	"6 \u2013 18"	✓
row:"Recommended range".unit	"modules"	"modules"	✓
row:"Recommended range".status	"ok"	"ok"	✓
status	"ok"	"ok"	✓
summary	"String size 6\u201318 modules (6 typical)"	"String size 6\u201318 modules (6 typical)"	✓
notes.count	2	2	✓
notes[0].standard	"IEC"	"IEC"	✓
notes[0].reference	"IEC 62548"	"IEC 62548"	✓
notes[0].text	"PV array design requirements. Includes temperature coeff..."	"PV array design requirements. Includes temperature coeff..."	✓
notes[1].standard	"IEEE"	"IEEE"	✓
notes[1].reference	"IEEE 1547"	"IEEE 1547"	✓
notes[1].text	"Interconnection requirements \u2014 no string sizing per..."	"Interconnection requirements \u2014 no string sizing per..."	✓
recommendations.count	2	2	✓

Field	Expected	Actual	Pass
recs[0]	"Validate Vmp > MPPT min under all expected operating con...	"Validate Vmp > MPPT min under all expected operating con...	✓
recs[1]	"Validate Voc (cold) < V_inverter_max to avoid over-volta...	"Validate Voc (cold) < V_inverter_max to avoid over-volta...	✓
formulas.count	4	4	✓
formulas[0].name	"Vmp (hot)"	"Vmp (hot)"	✓
formulas[0].expression	"Vmp_T = Vmp\u00b7(1 + \u03b2\u00b7(Tmax \u2212 25))"	"Vmp_T = Vmp\u00b7(1 + \u03b2\u00b7(Tmax \u2212 25))"	✓
formulas[0].variables	"\u03b2 \u2248 \u2212120.0035/\u00b0C"	"\u03b2 \u2248 \u2212120.0035/\u00b0C"	✓
formulas[1].name	"Voc (cold)"	"Voc (cold)"	✓
formulas[1].expression	"Voc_T = Voc\u00b7(1 + \u03b2\u00b7(Tmin \u2212 25))"	"Voc_T = Voc\u00b7(1 + \u03b2\u00b7(Tmin \u2212 25))"	✓
formulas[1].variables	""	""	✓
formulas[2].name	"N_min"	"N_min"	✓
formulas[2].expression	"N_min = \u2308Vmppt_min / Vmp_hot\u2309"	"N_min = \u2308Vmppt_min / Vmp_hot\u2309"	✓
formulas[2].variables	""	""	✓
formulas[3].name	"N_max"	"N_max"	✓
formulas[3].expression	"N_max = \u230aVinv_max / Voc_cold\u230b"	"N_max = \u230aVinv_max / Voc_cold\u230b"	✓
formulas[3].variables	""	""	✓
steps.count	">=0"	0	✓

HP ↔ kW Converter — hp-kw-converter

Test: 10 HP → kW · Category: conversion · Result: **PASS** (17/17 checks)

Input

```
{
  "value": 10,
  "from": "hp",
  "to": "kW"
}
```

Validation checks

Field	Expected	Actual	Pass
row:"Input".value	10	10	✓
row:"Input".unit	"hp"	"hp"	✓
row:"Output".value	7.457	7.457	✓
row:"Output".unit	"kW"	"kW"	✓
row:"Output".status	"ok"	"ok"	✓
status	"(none)"	"(none)"	✓
summary	"10 hp = 7.457 kW"	"10 hp = 7.457 kW"	✓
notes.count	0	0	✓
recommendations.count	0	0	✓
formulas.count	2	2	✓
formulas[0].name	"HP \u2192 W"	"HP \u2192 W"	✓
formulas[0].expression	"W = HP \u00b7 745.7"	"W = HP \u00b7 745.7"	✓
formulas[0].variables	"Mechanical"	"Mechanical"	✓
formulas[1].name	"PS \u2192 W"	"PS \u2192 W"	✓
formulas[1].expression	"W = PS \u00b7 735.5"	"W = PS \u00b7 735.5"	✓
formulas[1].variables	"Metric"	"Metric"	✓
steps.count	">=0"	0	✓

AWG ↔ mm² Converter — awg-mm2-converter

Test: AWG 4 → mm² · Category: conversion · Result: **PASS** (21/21 checks)

Input

```
{
  "mode": "awgTomm2",
  "value": 4
}
```

Validation checks

Field	Expected	Actual	Pass
row:"AWG / kcmil".value	"4"	"4"	✓
row:"AWG / kcmil".unit	""	""	✓
row:"mm²".value	21.1	21.1	✓
row:"mm²".unit	"mm\u00b2"	"mm\u00b2"	✓
row:"Cu ampacity (60 °C, NEC 310.16)".value	85	85	✓
row:"Cu ampacity (60 °C, NEC 310.16)".unit	"A"	"A"	✓
row:"Cu ampacity (60 °C, NEC 310.16)".status	"info"	"info"	✓
summary	"21.1 mm\u00b2 \u2248 4 AWG \u00b7 85 A @ 60 \u00b0C"	"21.1 mm\u00b2 \u2248 4 AWG \u00b7 85 A @ 60 \u00b0C"	✓
notes.count	1	1	✓
notes[0].standard	"NEC"	"NEC"	✓
notes[0].reference	"NEC 310.16"	"NEC 310.16"	✓
notes[0].text	"Allowable ampacity for 60/75/90 \u00b0C conductors, copp..."	"Allowable ampacity for 60/75/90 \u00b0C conductors, copp..."	✓
recommendations.count	0	0	✓
formulas.count	2	2	✓
formulas[0].name	"AWG \u2192 mm\u00b2"	"AWG \u2192 mm\u00b2"	✓
formulas[0].expression	"mm\u00b2 \u2248 0.012668 \u00b7 92^ ((36\u2212n)/19.5)"	"mm\u00b2 \u2248 0.012668 \u00b7 92^ ((36\u2212n)/19.5)"	✓
formulas[0].variables	"n = AWG number"	"n = AWG number"	✓
formulas[1].name	"kcmil \u2192 mm\u00b2"	"kcmil \u2192 mm\u00b2"	✓
formulas[1].expression	"mm\u00b2 = kcmil / 1.9735"	"mm\u00b2 = kcmil / 1.9735"	✓
formulas[1].variables	""	""	✓
steps.count	">=0"	0	✓